NLM and Friction Revision Sheet

Bonus Questions **Time to be Taken: 20 Min Max**

1. Column II gives certain situations involving two blocks of mass 2 kg and 4 kg. The 4 kg block lies on a smooth horizontal table. There is sufficient friction between both the block and there is no relative motion between both the blocks in all situations. Horizontal forces act on one or both blocks as shown. Column I gives certain statement related to figures given in column II. Match the statements in column I with the figure in column II.

Column I

(A) Magnitude of frictional force is maximum.

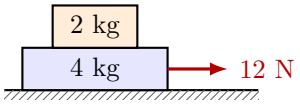
(B) Magnitude of friction force is least.

(C) Friction force on 2 kg block is towards right.

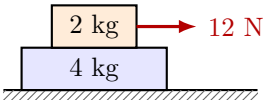
(D) Friction force on 2 kg block is towards left.

Column II

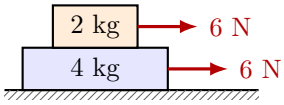
(p)



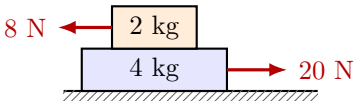
(q)



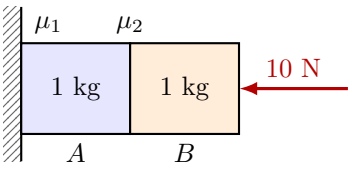
(r)



(s)



2. In the given figure find the accelerations of blocks A and B for the following cases ($g = 10$ m/s^2)



Column - I

(A) $\mu_1 = 0$ and $\mu_2 = 0.1$

(B) $\mu_2 = 0$ and $\mu_1 = 0.1$

(C) $\mu_1 = 0.1$ and $\mu_2 = 1.0$

(D) $\mu_1 = 1.0$ and $\mu_2 = 0.1$

Column - II

(p) $a_A = a_B = 9.5$ m/s^2

(q) $a_A = 9$ m/s^2, $a_B = 10$ m/s^2

(r) $a_A = a_B = g = 10$ m/s^2

(s) $a_A = 1,\ a_B = 9$ m/s^2

Keep Going!

"Champions are made in the extra effort nobody asks for. Take this bonus round and finish strong."